StyleMorph: Learning a Morphable 3D Model from 2D Photos Alone

ICLR 2023 · UCL and the Alan Turing Institute give a StyleGAN clean, independent dials for shape, pose, and appearance, learned from ordinary 2D photos with no 3D scans and no template built by hand.

Modern 3D-aware GANs make gorgeous, view-consistent portraits. What they do not make is editable ones. Because their latent codes fuse geometry and appearance into a single hybrid representation, you cannot rotate a generated face without also nudging its identity, or swap the background without disturbing the subject. The pixels are beautiful; the control knobs are glued together.

The graphics world solved the control problem decades ago with 3D Morphable Models (3DMMs), the workhorse behind visual-effects pipelines and AR avatars. A 3DMM hands the artist clean, separate dials for pose, expression, and appearance. The catch is how you build one: classical 3DMMs demand 3D scanning rigs and painstaking manual alignment across a subject population, which is exactly why they mostly exist for a single category, the human face.

StyleMorph, from University College London and the Alan Turing Institute, asks whether we can get morphable-model control inside a modern generator while learning everything from nothing but unstructured 2D photos. The answer is a model that cleanly pulls apart four factors of an image, the object's 3D shape, the camera pose, the foreground appearance, and the background, and lets you edit each one independently, all while matching the image quality of the strongest 3D-aware GANs that offer none of this control.

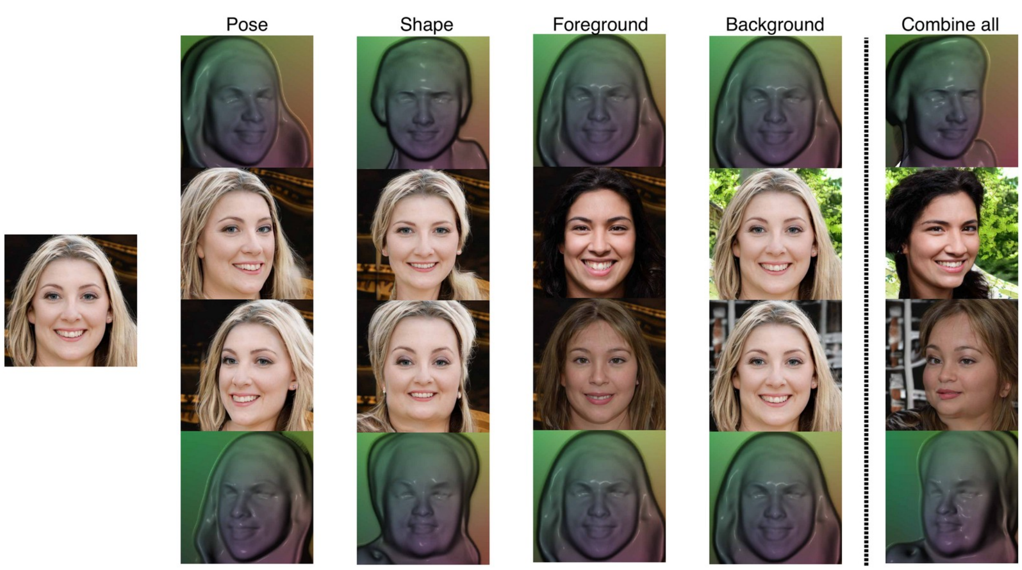

Figure 1. Disentangled control in StyleMorph. Starting from one synthesized sample (far left), each column changes a single factor: pose, shape, foreground appearance, or background. The rightmost column combines all four. The geometric conditioning signal (top and bottom rows) stays fixed across the foreground and background edits, so those two changes come purely from the appearance codes.

Paper: https://openreview.net/forum?id=Ojpb1y8jflw · Project page: https://stylemorph.github.io/stylemorph/

Why 3D-aware GANs entangle, and what a morphable model buys you

The recent wave of 3D-aware GANs, from GIRAFFE to EG3D, earns its realism by baking a 3D inductive bias into the generator: a neural radiance field or an implicit occupancy network renders the scene volumetrically, so different viewpoints stay consistent by construction. The price is entanglement. A single style vector, or a shared feature volume, drives shape and texture together, so there is no place to reach in and change one factor without touching the others. Some hybrids add partial disentanglement, but none of them learn a reusable, category-general shape model that you can morph at will.

A morphable model is exactly that missing piece: a canonical template plus a learned space of deformations that warp it into each individual instance. If you could obtain one for arbitrary categories without 3D supervision, you would get both worlds, the photorealism of a GAN and the surgical editability of a 3DMM. StyleMorph's contribution is to make the morphable model itself something the network discovers by backpropagation, from 2D images alone.

The idea: funnel all geometry through one purely geometric bottleneck

StyleMorph has two halves that meet at a single, deliberately narrow interface. The first half is a Morphable Renderer that handles geometry. It learns a deformation-free object as a constant implicit field living in a canonical template space. A deformation field built on SIREN, driven by a shape code, warps each camera ray from world space into that template space by adding a predicted offset, r-hat(t) = r(t) + g_s(r(t)). Integrating the template coordinates along each ray produces a 2D map the authors call a TOCS map, for Template Object Coordinates.

TOCS is the conceptual heart of the paper. It is a deformable cousin of Normalized Object Coordinates: every surface point carries a stable identity in template space, so the map is a deformation-equivariant descriptor of 3D shape. Crucially, a TOCS map is exclusively geometric. It encodes the compounded effect of shape deformation, camera pose, and perspective projection, and nothing at all about texture or color. That is the bottleneck: by forcing every geometric factor through this one 2D image, and nothing else through it, StyleMorph guarantees that appearance has to enter the pipeline somewhere else.

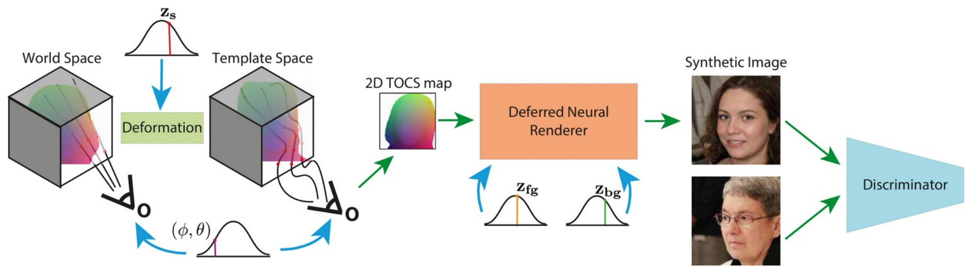

Figure 2. The two halves of StyleMorph. Left: a template shape, a shape code z_s, camera pose (phi, theta), and perspective projection are combined into a single 2D TOCS map that bottlenecks all geometric variation. Right: the TOCS map, plus separate foreground (z_fg) and background (z_bg) appearance codes, drives a Deferred Neural Renderer, trained against a discriminator, to produce a high-resolution image.

The second half is where appearance lives. A Deferred Neural Renderer (DNR) built on StyleGAN2 takes the geometric TOCS map together with two independent appearance codes, one for the foreground object and one for the background scene, and synthesizes a high-resolution image. Foreground and background are generated separately and composited late, through a predicted alpha mask, I = A * I_fg + (1 - A) * I_bg. Because shape and pose can only reach the renderer through the TOCS map, and the two appearance codes never touch geometry, the four factors come out cleanly separated by design rather than by a regularization penalty.

Training runs in two stages. In stage one, the deformable volume renderer is trained as a low-resolution 64-squared RGBA image generator, so it learns a realistic shape model, with only weak silhouette supervision from off-the-shelf Labels4free masks. In stage two, that renderer is frozen and its RGB block pruned; it now serves only to emit 64-squared TOCS maps, while the DNR is trained at full resolution to turn those maps plus appearance codes into photorealistic output. Splitting the work this way lets the expensive volumetric model stay small and cheap, while the 2D renderer carries the resolution.

Does the disentanglement actually hold?

The first test is whether the learned shape space is rich and whether the renderer respects it. Fixing the viewpoint and appearance codes and varying only the shape code should change face and hair geometry while leaving the rendered texture aligned to the new shape.

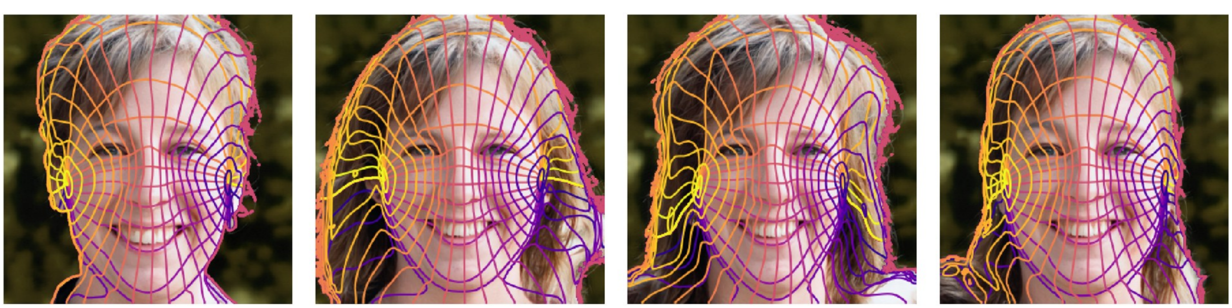

Figure 5. Shape control from a fixed viewpoint. Four different shape codes yield four different TOCS maps (varying hair styles and face shapes), and the rendered RGB tracks them closely. Equicontours drawn on the template coordinates reveal dense surface correspondences that hold across shapes.

The equicontours make the point precise: the same template coordinate lands on the same semantic location across very different faces, which is what a genuine morphable model is supposed to guarantee. The next test is the full four-way edit, and whether it generalizes beyond faces to categories a classical 3DMM never covered.

Figure 6. Independent editing of one factor at a time, across faces, cats, and wild animals. From the source (yellow border), StyleMorph varies shape (green), foreground appearance (blue), and background (red) while holding the other factors fixed.

Changing the foreground code restyles the subject without moving the geometry; changing the background code repaints the scene without touching the subject; changing the shape code alters the underlying 3D form. The same behavior holds on cats and wild animals, evidence that StyleMorph has effectively learned an unsupervised morphable model for each category, not just for faces.

Disentanglement without paying for it in image quality

The headline empirical result is that all of this control is essentially free in FID. On FFHQ faces at 256-squared, StyleMorph reaches an FID of 7.91, sitting comfortably among the strongest 3D-aware GANs. The comparison that matters most is against Disentangled3D, the only prior method that also represents shape with a deformable template: StyleMorph's 7.91 is far ahead of its 28.18, and StyleMorph additionally separates foreground from background, which that method does not.

Table 1. FID at 256-squared on FFHQ and AFHQ (lower is better). StyleMorph is competitive with the best hybrids while adding full four-way disentanglement and a learned deformable template. A dash means the paper does not report that split. EG3D uses weak 3D supervision (a pre-trained pose estimator) and offers no factor disentanglement; Disentangled3D is the only other method with a deformable template.

Method	FFHQ	AFHQ Cats	AFHQ Wild	AFHQ Dogs
StyleMorph (ours)	7.91	4.29	3.49	13.95
Disentangled3D	28.18	-	-	-
GiraffeHD	11.93	12.36	-	-
EG3D (weak 3D sup.)	4.80	3.88	-	-

On animals the story repeats: 4.29 on Cats, 3.49 on Wild, and 13.95 on Dogs. EG3D posts lower raw FID on faces, but it leans on a pre-trained pose estimator for weak 3D supervision and offers none of StyleMorph's factor control, so the two are not doing the same job. The point of the table is not to top every column, but to show that adding a learned template and four-way editability does not force a quality tax.

Editing real photographs, not just samples

Disentangled control is most useful when you can point it at a real image. By inverting the pipeline, StyleMorph recovers the latent codes and underlying 3D structure of a genuine input photo, and then re-renders it under new pose, foreground, background, or shape, each dial turned independently.

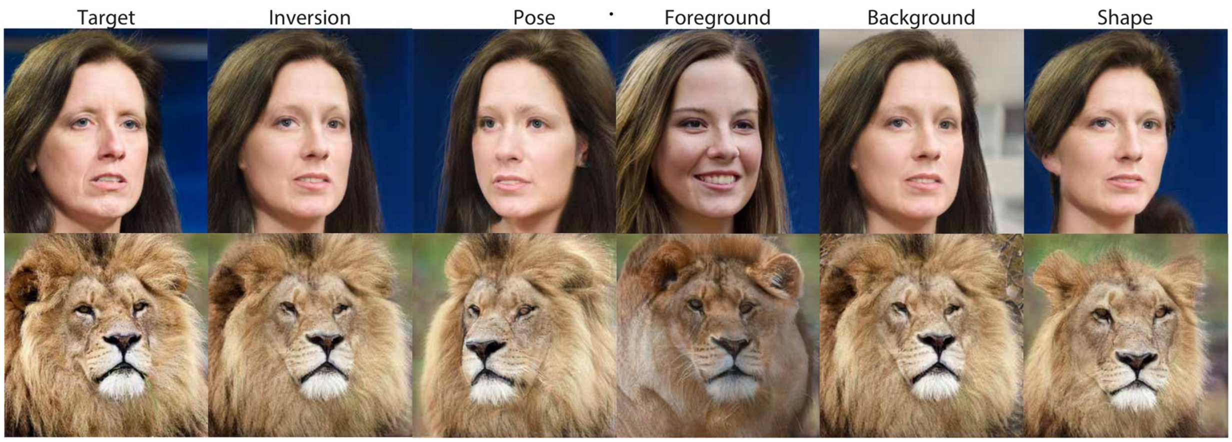

Figure 8. Editing a real photo by inversion. StyleMorph inverts a target image (left) into its latent codes and recovers the underlying 3D structure, then re-renders it under new pose, foreground, background, and shape, each controlled independently.

What the ablations confirm

Under a fair 96-hour training budget, the ablations isolate each design choice. Swapping TOCS for plain NOCS, while keeping everything else identical, worsens FID from 8.31 to 8.90 and visibly blurs the conditioning signal, confirming that template-space coordinates, not just object coordinates, are what sharpen both quality and disentanglement. Removing the deformable module in favor of a direct occupancy network keeps FID similar but wrecks shape and appearance consistency. And switching from late fusion to early fusion nudges FID down slightly but hurts alpha consistency (88.65% to 85.87%), letting foreground and background bleed together, which is why StyleMorph keeps them separate until the final composite.

There is an honest trade-off worth naming. Directly optimizing a differentiable view-consistency loss does improve that metric substantially (from 15.84 to 8.12), but it costs image quality, pushing FID up to 12.31. StyleMorph's default configuration chooses photorealism over squeezing out the last of view consistency, a reasonable call, but one the paper is transparent about rather than hiding.

Takeaway

StyleMorph delivers morphable-model control inside a state-of-the-art image generator, learned entirely from ordinary 2D photos with no 3D scans, pose labels, or predefined template.

By morphing a learned canonical template and rendering it into a purely geometric TOCS map, it hands a StyleGAN renderer a clean geometric signal, so shape, pose, foreground appearance, and background can each be edited on their own without sacrificing photorealism.

The broader significance is that StyleMorph effectively builds an unsupervised 3D morphable model for general object categories. Where classical morphable models required scanning rigs and existed almost exclusively for human faces, here the template and its space of deformations are discovered by backpropagation from an unlabeled image collection. That extends the fine-grained controllability visual-effects artists have long relied on from faces to cats, dogs, and wild animals, and it puts editable, category-general 3D-aware synthesis within reach of anyone with a folder of photos rather than a capture studio.

The boundaries are the usual ones for this line of work. Results are demonstrated on single-object, roughly centered categories, and the ablations are honest about a measurable quality-versus-view-consistency trade-off rather than a free lunch on both. But as a demonstration that disentanglement and photorealism can coexist, and that a morphable model can be learned rather than hand-built, StyleMorph is a clean and genuinely useful step toward controllable generative modelling.